

DTMC

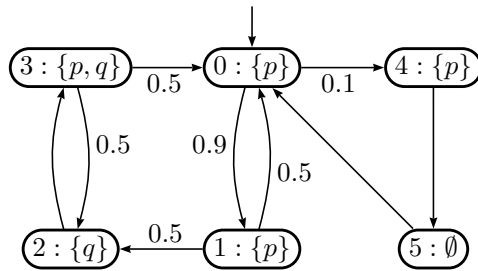
Chance

1. PCTL formulas are ϕ with the following syntax:

$$\begin{aligned}\phi &::= \mathbf{true} \mid p \mid \phi_1 \wedge \phi_2 \mid \neg\phi \mid \mathbf{P}_{\sim p}[\psi] && \text{(state formulas)} \\ \psi &::= \mathbf{X}\phi \mid \phi_1 \mathbf{U}^{\leq k} \phi_2 \mid \phi_1 \mathbf{U} \phi_2 && \text{(path formulas)}\end{aligned}$$

where p is an atomic proposition, and $\sim$ is one of $\leq, <, \geq, >, =$.

- (a) Give the formal semantic of PCTL, interpreted over DTMC.
 - (b) Let's introduce some derived operators with the 'usual' meaning: $\forall, \rightarrow$, and $\mathbf{F}$ (or $\Diamond$). Propose a definition for them.
 - (c) Developer Peter Parker claims that $\neg\mathbf{P}_{>c}\phi$ is equivalent to saying $\mathbf{P}_{\leq c}\phi$. Is Peter right?
 - (d) The 'usual' way to define $\mathbf{G}$ is: $\mathbf{G}\phi = \neg\mathbf{F}\neg\phi$. So, for example $\mathbf{P}_{\geq 0.8}[\mathbf{G}a]$ can be expressed as $\mathbf{P}_{\geq 0.8}[\neg\mathbf{F}\neg a]$. Unfortunately, the syntax of PCTL does not have negation over the temporal formula (we only have negation over proposition and probabilistic formulas). So, the formula " $\mathbf{P}_{\geq 0.8}[\neg\mathbf{F}\neg a]$ " is not allowed in PCTL. How do you then propose to define $\mathbf{P}_{\geq p}[\mathbf{G}\phi]$?
2. Explain why in PCTL (e.g. interpreted over DTMC) $\mathbf{P}_{\geq 1}[\mathbf{F}\phi]$ is not the same as CTL $\mathbf{A}\mathbf{F}\phi$ (the former is weaker than the latter). Give an example that shows the difference. Note: $\mathbf{F}\phi$ is defined as $\mathbf{true} \mathbf{U} \phi$.
 3. Consider the following DTMC M modelling some system. $Prop = \{p, q\}$.



Consider the following PCTL properties to check:

- $\mathbf{P}_{\geq 0.82} [p \mathbf{U} q]$
- $\mathbf{P}_{< 0.82} [p \mathbf{U} q]$
- $\mathbf{P}_{=?} [p \mathbf{U} q]$

What do these formula say? Then describe the procedure to check them (or at least one of them) on M .

4. Consider again the DTMC M in No. 3. On which states of M the following properties hold?

- $\mathbf{P}_{\leq 0} [p \mathbf{U} \neg(p \vee q)]$.

- $\mathbf{P}_{\geq 1} [p \mathbf{U} \neg(p \vee q)]$.
- $\mathbf{P}_{\geq 0.18} [p \mathbf{U} \neg(p \vee q)]$.
- $\mathbf{P}_{< 0.18} [p \mathbf{U} \neg(p \vee q)]$.

5. Consider again the DTMC M in No. 3. Which states satisfy $\mathbf{P}_{\leq 0.5} [p \mathbf{U} \mathbf{P}_{\geq 1}[q \mathbf{U} p]]$? (Well, obviously state 5, but which other states, if any, satisfy the property?)
6. PLTL formulas ϕ has the following syntax: where p is an atomic proposition, and $\sim$ is one of $\leq, <, \geq, >, =$.

$$\begin{aligned} \phi &::= \mathbf{P}_{\sim p}[\psi] \\ \psi &::= p \mid \psi_1 \wedge \psi_2 \mid \neg\psi \mid \mathbf{X}\psi \mid \psi_1 \mathbf{U} \psi_2 \quad (\text{path formulas}) \end{aligned}$$

Note that a path formula can only recurse over other path formulas. In particular, unlike in PCTL, a PLTL path formula cannot have a probabilistic sub-formula.

Give the formal semantic of PLTL, interpreted over DTMC.

7. Given an example that demonstrates that PLTL formula $\mathbf{P}_{\geq 1}[F\psi]$ is not the same as plain LTL formula $\Diamond\psi$.
8. Consider again the DTMC M in No. 3.
- (a) What are the SCCs of M ? Which ones are bottom SCCs (BSCCs)?
 - (b) Does the PLTL property $\mathbf{P}_{\geq 1.0}[\mathbf{GF}(p \wedge q)]$ hold on M ?
 - (c) How about $\mathbf{P}_{> 0}[\mathbf{FG}p]$?

9. Consider again the DTMC M in No. 3. We want to check if it satisfies the PLTL properties:

- $\mathbf{P}_{\leq 0.8}[(p \wedge \neg q) \mathbf{U} \mathbf{G}q]$.
- $\mathbf{P}_{\leq 0.8}[p \mathbf{U} \mathbf{G}q]$.

Perform the PLTL model checking algorithm to check them.